

AKurIT-4

RIFAMPIN, ISONIAZID, PYRAZINAMIDE AND ETHAMBUTOL HYDROCHLORIDE TABLETS U.S.P

Akurit-4 is a four drug Fixed Dose Combination (FDC) of anti-TB drugs Rifampin, Isoniazid, Pyrazinamide and Ethambutol developed as per the recommendations of WHO, which offers the ease of administrations and facilitates patient compliance in the treatment of tuberculosis.

Composition:

Each film coated tablet contains:

Rifampin USP	150 mg
Isoniazid USP	75 mg
Pyrazinamide USP	400 mg
Ethambutol HCL USP	275 mg

Product Description:

Brown colored, capsule shaped film-coated tablets with break line on one side and plain on the other side.

CLINICAL PHARMACOLOGY

PHARMACODYNAMICS

Antibacterial Activity

Rifampin

Escherichia coli, Pseudomonas, Indole negative and indole positive Proteus, Klebsiella, Staphylococcus aureus, Coagulase-negative staphylococci, Neisseria meningitides, Haemophilus influenza, Legionella species, M tuberculosis, M kansasii, M scrofulaceum, M intracellulare & M avium.

Isoniazid

It exerts an active bactericidal action on M tuberculosis which it inhibits in concentrations of 0.05-2 mg/l. Other mycobacteria are generally resistant, although some strains of M kansasii are susceptible.

Ethambutol

M tuberculosis, M fortuitum, M kansasii, M intracellulare, M avium, M ulcerans (some strains) & M marinum (some strains)

Pyrazinamide

Its activity against M. tuberculosis is highly pH dependent. It is almost inactive at neutral pH; at pH 5.5 the MIC is 16-32 mg/l. Other Mycobacteria are resistant. Its action, which depends on intracellular conversion to pyrazinoic acid in M. tuberculosis but evidently not in other mycobacteria, is bactericidal.

PHARMACOKINETICS

Rifampin

The oral administration of rifampin produces peak concentrations in plasma in 2 to 4 hours. The half-life of rifampin varies from 1.5 to 5 hours and is increased in the presence of hepatic dysfunction; it may be decreased in patients receiving isoniazid concurrently who are slow inactivators of this drug. Up to 30% of a dose of rifampin is excreted in the urine; less than half of this may be unaltered antibiotic. Adjustment of dosage is not necessary in patients with impaired renal function.

Isoniazid

Peak plasma concentrations of 3 to 5 mcg/ml develop 1 to 2 hours after oral ingestion of usual doses. From 75 to 95% of a dose of isoniazid is excreted in the urine within 24 hours, as metabolites. The main excretory products in man are the results of enzymatic acetylation (acetylisoniazid) and enzymatic hydrolysis (isonicotinic acid). The rate of acetylation significantly alters the concentrations of the drug that are achieved in plasma and its half-life in the circulation. The half-life of the drug may be prolonged in the presence of hepatic insufficiency.

Ethambutol

After oral administration, 75 to 80% of ethambutol is absorbed from the gastrointestinal tract. A single dose of 15 mg/kg produces a plasma concentration of about 5 mcg/ml at 2 to 4 hours. The drug has a half life of 3 to 4 hours. Within 24 hours, two thirds of an ingested dose of ethambutol is excreted unchanged in the urine; upto 15% is excreted in the form of two metabolites, an aldehyde and a dicarboxylic acid derivative. Renal clearance of ethambutol is approximately 7 ml. min⁻¹ . kg⁻¹, and the drug is excreted by tubular secretion in addition to glomerular filtration.

Pyrazinamide

Pyrazinamide is well absorbed from the gastrointestinal tract and attains peak plasma concentrations within 2 hours. Plasma concentrations generally range from 30 to 50mcg/ml with doses of 20 to 25 mg/kg. It is widely distributed in body tissues and fluids including the liver, lungs, and cerebrospinal fluid. It is approximately 10% bound to plasma proteins. The plasma half-life of pyrazinamide is 9 to 10 hours in patients with normal renal and hepatic function. The half-life of the drug may be prolonged in patients with impaired renal or hepatic function. Within 24 hours, approximately 70% of an oral dose of pyrazinamide is excreted in urine, mainly by glomerular filtration. About 4% to 14% of the dose is excreted as unchanged drug; the remainder is excreted as metabolites.

INDICATIONS

Akurit-4 is indicated in the treatment of both pulmonary and extra pulmonary tuberculosis, in the intensive initial phase of treatment. During initial phase that can last 3 months, Akurit-4 is administered on a daily continuous basis.

DOSAGE & ADMINISTRATION

Adults – 3 tablets per day for an average body weight of 50 kg.

This tablet can be broken into two for easy swallowing purpose. Not for taking half tablet dosage.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS:

Rifampin

Rifampin is contraindicated in patients with a history of hypersensitivity to any of the rifamycins.

Isoniazid

Previous isoniazid-associated hepatic injury; severe adverse reactions to isoniazid, such as fever, chills and arthritis; acute liver disease of any etiology, a history of previous hypersensitivity reaction to isoniazid, including drug induced hepatitis.

Ethambutol

Ethambutol is contraindicated in patients who are known to be hypersensitive to this drug. It is

contraindicated in patients with known optic neuritis unless clinical judgement determines that it may be used.

Pyrazinamide

Pyrazinamide is contraindicated in patients who are known to be hypersensitive to pyrazinamide.

WARNINGS AND PRECAUTIONS

Rifampin

Rifampin is not recommended for intermittent therapy; the patient should be cautioned against intentional or accidental interruption of the daily dosage regimen since rare renal hypersensitivity reactions have been reported when therapy was resumed in such cases.

Rifampin has been observed to increase the requirements for anticoagulant drugs of the coumarin type. The cause of the phenomenon is unknown. In patients receiving anticoagulants and Rifampin concurrently, it is recommended that the prothrombin time be performed daily or as frequently as necessary to establish and maintain the required dose of anticoagulant.

Urine, feces, saliva, sputum, sweat and tears may be colored red-orange by Rifampin and its metabolites. Soft contact lenses may be permanently stained. Individuals to be treated should be made aware of these possibilities.

It has been reported that the reliability of oral contraceptives may be affected in some patients being treated for tuberculosis with Rifampin in combination with at least one other antituberculosis drug. In such cases, alternative contraceptive measures may need to be considered.

It has also been reported that Rifampin given in combination with other antituberculosis drugs may decrease the pharmacologic activity of methasone, oral hypoglycemics, digloxin, quinidine, disopyramide, dapsone and corticosteroids. In these cases, dosage adjustment of the interacting drugs is recommended.

Therapeutic levels of Rifampin have been shown to inhibit standard microbiological assays for serum folate and vitamin B12. Alternative methods must be considered when determining folate and vitamin B12 concentrations in the presence of Rifampin. Since Rifampin has been reported to cross the placental barrier and appear in cord blood and in maternal milk, neonates and newborns of Rifampin-treated mothers should be carefully observed for any evidence of untoward effects.

Isoniazid

All drugs should be stopped and an evaluation of the patients should be made at the first sign of a hypersensitivity reaction.

Use of isoniazid should be carefully monitored in the following:

1. Patients who are receiving phenytoin (diphenylhydantoin) concurrently. Isoniazid may decrease the excretion of phenytoin or may enhance its effects. To avoid phenytoin intoxication, appropriate adjustment of the anticonvulsant dose should be made.
2. Daily users of alcohol. Daily ingestion of alcohol may be associated with a higher incidence of isoniazid hepatitis.
3. Patient with current chronic liver disease or severe renal dysfunction.

Periodic ophthalmoscopic examination during isoniazid therapy is recommended when visual symptoms occur.

Ethambutol

The effects of combinations of ethambutol with other antituberculous drugs on the fetus is not known. While administration of this drug to pregnant human patients has produced no detectable effect upon the fetus, the possible teratogenic potential in women capable of bearing children would be weighed carefully against the benefits of therapy. There are published reports of five women who received the drug during pregnancy without apparent adverse effect upon the fetus.

Ethambutol is not recommended for use in children under 13 years of age since safe conditions for use have not been established.

Patients with decreased renal function need the dosage reduced as determined by serum levels of ethambutol, since the main path of excretion of this drug is by the kidneys.

Because this drug may have adverse effects on vision, physical examination should include ophthalmoscopy, finger perimetry, and testing of colour discrimination. In patients with visual defects such as cataracts, recurrent inflammatory conditions of the eye, optic neuritis, and diabetic retinopathy, the evaluation of changes in visual acuity is more difficult, and care should be taken to be sure the variations in vision are not due to the underlying disease conditions. In such patients expected and possible visual deterioration since evaluation of visual changes is difficult. As with any potent drug, periodic assessment of organ system functions, including renal, hepatic, and hematopoietic, should be made during long-term therapy.

Pyrazinamide

Pyrazinamide inhibits renal excretion of urates, frequently resulting in hyperuricemia which is usually asymptomatic. Pyrazinamide also cause hyperuricemia which is accompanied by acute gouty arthritis.

Patients should have baseline serum uric acid and liver function determinations. Patients with preexisting liver disease or those patients at increased risk for drug related hepatitis (e.g. alcohol abusers) should be followed closely.

Because it contains pyrazinamide, AKurIT-4 should be discontinued and not be resumed if signs of hepatocellular damage or hyperuricemia accompanied by an acute gouty arthritis appear. If hyperuricemia accompanied by an acute gouty arthritis occurs without liver dysfunction, the patient should be transferred to a regimen not containing pyrazinamide.

INTERACTIONS WITH OTHER MEDICINES

Rifampin

Enzyme Induction: Rifampin is known to induce certain cytochrome P-450 enzymes.

Rifampin has been reported to accelerate the metabolism of the following drugs: anticonvulsants (eg. phenytoin), antiarrhythmics (eg. disopyramide, mexiletine, quinidine, tocainide), anticoagulants, antifungal (eg. fluconazole, itraconazole, ketoconazole), barbiturates, beta-blockers, calcium channel blockers (eg. diltiazem, nifedipine, verapamil), chloramphenicol, ciprofloxacin, corticosteroids, cyclosporine, cardiac glycoside preparations, clofibrate, oral contraceptives, dapsone, diazepam, haloperidol, oral hypoglycemic agents (sulfonylureas, methadone, narcotic analgesics, nortriptyline, progestins, and theophylline).

Rifampin has been observed to increase the requirements for anticoagulant drugs of the coumarin type. In patients receiving anticoagulants and Rifampin concurrently, it is recommended that the prothrombin time be performed daily or as frequently as necessary to establish and maintain the required dose of anticoagulant.

Concurrent use of ketoconazole and Rifampin has resulted in decreased serum concentration of both drugs. Concurrent use of Rifampin and enalapril has resulted in decreased concentrations of enalaprilat, the active metabolite of enalapril.

Concomitant antacid administration may reduce the absorption of Rifampin. Daily doses of AKurIT-4, because it contains rifampin, should be given at least 1 hour before the ingestion of antacids.

Probencid and cotrimoxazole have been reported to increase the blood level of rifampin. When rifampin is given concomitantly with either halothane or isoniazid the potential for hepatotoxicity is increased.

Isoniazid

Enzyme Inhibition: Isoniazid is known to inhibit certain cytochrome P-450 enzymes. Co administration of isoniazid with drugs that undergo biotransformation through these metabolic pathways may decrease elimination. Consequently, dosages of drugs metabolized by these enzymes may require adjustment when starting or stopping concomitantly administered AKurIT-4, because it contains isoniazid, to maintain optimum therapeutic blood levels.

Isoniazid has been reported to inhibit the metabolism of the following drugs:

Anticonvulsants (eg. Carbamazepine, phenytoin, primidone, valproic acid), benzodiazepines (eg. diazepam), haloperidol, ketoconazole, theophylline, and warfarin. It may be necessary to adjust the

dosage of these drugs if they are given concurrently with RIFATER because it contains isoniazid. The impact of the competing effects of rifampin and isoniazid on the metabolism of these drugs is unknown.

Ethambutol

Co-administration of ethambutol and ethionamide may intensify adverse effects of ethambutol in patients receiving this combination of agents. Adverse reactions may include jaundice, hepatitis, neuritis, vision problems, headache, dizziness, confusion, or GI distress.

Pyrazinamide

Pyrazinamide may induce cyclosporine metabolism resulting in reduced serum concentrations of cyclosporine and potential loss of immunosuppressive efficacy. Co administration of ethionamide and pyrazinamide may potentiate hepatotoxicity in patients receiving this combination of agents. Fosphenytoin is prodrug of phenytoin and the same interactions that occur with phenytoin are expected to occur with fosphenytoin. Synergism between isoniazid and pyrazinamide that may have resulted in an increase in phenytoin serum concentrations and phenytoin toxicity. Pyrazinamide may decrease the effect of probenecid on uric acid excretion. This may result in increased serum uric acid levels and precipitation or aggravation of symptoms in patients with gout. Zidovudine was shown to significantly reduce serum concentrations of pyrazinamide in a pilot study involving patients with AIDS and tuberculosis.

PREGNANCY AND LACTATION:

Rifampicin

Rifampicin has been shown to be teratogenic in rodents given oral doses of rifampicin 15 to 25 times the human dose. Although rifampicin has been reported to cross the placental barrier and appear in cord blood, the effect of rifampicin alone or in combination with other anti-tuberculosis drugs, on human foetus is not known.

Pyrazinamide

Animal reproduction studies have not been conducted with pyrazinamide. It is also not known whether pyrazinamide can cause foetal harm when administered to pregnant women or can affect reproduction capacity. Pyrazinamide should be given to pregnant women only if clearly needed

Ethambutol

In foetuses born of mice treated with high doses of Ethambutol during pregnancy, a low incidence of cleft palate, exencephaly and abnormality of the vertebral column was noted.

An extensive review of the literature of its use during pregnancy led to the conclusion that it was second only to isoniazid in its safety, and that the rate of abnormalities reported was low and that most of them were of a minor or thopaedic nature, not necessarily related to drug administration.

Isoniazid

It has been reported that in both rats and rabbits, isoniazid may exert an embryocidal effect when administered orally during pregnancy, although no isoniazid related congenital anomalies have been found in reproduction studies in mammalian species (mice, rats and rabbits). Isoniazid should be prescribed during pregnancy only when therapeutically necessary.

SIDE EFFECTS:

Rifampin

NERVOUS SYSTEM REACTIONS:

Headache, drowsiness, fatigue, ataxia, dizziness, inability to concentrate, mental confusion, visual disturbances, muscular weakness, pain in extremities and generalized numbness.

GASTROINTESTINAL DISTURBANCES:

In some patients heartburn, epigastric distress, anorexia, nausea, vomiting, gas, cramps, and diarrhea.

HEPATIC REACTIONS:

Transient abnormalities in liver function tests (e.g., elevations in serum bilirubin, BSP, alkaline phosphatase, serum transaminases) have been observed. Rarely, hepatitis or a shocklike syndrome with hepatic involvement and abnormal liver function tests.

RENAL REACTIONS:

Elevations in BUN and serum uric acid have been reported. Rarely, hemolysis, hemoglobinuria, hematuria, interstitial nephritis, renal insufficiency and acute renal failure have been noted. These are generally considered to be hypersensitivity reactions. They usually occur during intermittent therapy or when treatment is resumed following intentional or accidental interruption of a daily dosage regimen, and are reversible when rifampin is discontinued and appropriate therapy instituted.

HEMATOLOGIC REACTIONS:

Thrombocytopenia, transient leucopenia, hemolytic anemia, eosinophilia and decreased hemoglobin have been observed. Thrombocytopenia has occurred when rifampin and ethambutol were administered concomitantly according to an intermittent dose schedule twice weekly and in high doses.

ALLERGIC AND IMMUNOLOGICAL REACTIONS :

Occasionally pruritis, urticaria, rash, pemphigoid reaction, eosinophilia, sore mouth, sore tongue, and exudate conjunctivitis. Rarely hemolysis, hemoglobinuria, hematuria, renal insufficiency or acute renal failure have been reported which are generally considered to be hypersensitivity reactions. These have usually occurred during intermittent therapy or when treatment was resumed following intentional or accidental interruption of a daily dosage regimen and were reversible when rifampin was discontinued and appropriate therapy instituted.

METABOLIC REACTIONS:

Elevations in BUN and serum uric acid have occurred.

MISCELLANEOUS REACTIONS:

Fever and menstrual disturbances have been noted.

Isoniazid

NERVOUS SYSTEM REACTIONS:

Peripheral neuropathy is the most common toxic effect. It is dose-related, occurs most often in the malnourished and in those predisposed to neuritis (e.g., alcoholics and diabetics), and is usually preceded by paresthesias of the feet and hands. The incidence is higher in "slow inactivators". Other neurotoxic effects, which are uncommon with conventional doses, are convulsions, toxic encephalopathy, optic neuritis and atrophy, memory impairment, and toxic psychosis.

GASTROINTESTINAL REACTIONS:

nausea, vomiting, epigastric distress and pancreatitis.

HEPATIC REACTIONS:

Elevated serum transaminases (SGOT; SGPT), bilirubinemia, bilirubinuria, jaundice and occasionally severe and sometimes fatal hepatitis. The common prodromal symptoms are anorexia, nausea, vomiting, fatigue, malaise, and weaknesses. Mild and transient elevations of serum transaminase levels occurs in 10 to 2% of persons taking isoniazid. The abnormality usually occurs in the first 4 to 6 months of treatment but can occur at any time during therapy. In most instances, enzyme levels return to normal with no necessity to discontinue medication. In occasional instances,

progressive liver damage occurs, with accompanying symptoms. In these cases, the drug should be discontinued immediately. The frequency of progressive liver damage increases with age. It is rare in persons under 20, but occurs in up to 2.3% of those over 50 years of age.

HEMATOLOGIC REACTIONS:

Agranulocytosis, hemolytic sideroblastic or aplastic anemia, thrombocytopenia and Eosinophilia.

HYPERSENSITIVITY REACTIONS:

Fever, skin eruptions (morbilliform, maculopapular, purpuric, or exfoliative), lymphadenopathy and vasculitis.

METABOLIC AND ENDOCRINE REACTIONS:

Pyridoxine deficiency, pellagra, hyperglycemia, metabolic acidosis, and gynecomastia.

MISCELLANEOUS REACTIONS:

Rheumatic syndrome and systemic lupus erythematosus-like syndrome.

Ethambutol

Ethambutol may produce decreases in visual acuity which appear to be due to optic neuritis and to be related to dose and duration of treatment. The effects are generally reversible when administration of the drug is discontinued promptly. In rare cases recovery may be delayed for up to 1 year or more and the effect may possibly be irreversible in these cases. The change in visual acuity may be unilateral or bilateral and hence each eye must be tested separately and both eyes tested together. Testing of visual acuity should be performed before beginning ethambutol therapy and periodically during drug administration, except that it should be done monthly when a patient is on a dosage of more than 15 mg/kg per day. If careful evaluation confirms the magnitude of visual change and fails to reveal another cause, Ethambutol should be discontinued and the patient reevaluated at frequent intervals. Progressive decreases in visual acuity during therapy must be considered to be due to ethambutol hydrochloride.

Ethambutol may show subjective visual symptoms before, or simultaneously with, the demonstration of decreases in visual acuity, and all patients receiving ethambutol should be questioned periodically about blurred vision and other subjective eye symptoms.

Other adverse reactions reported include: anaphylactoid reactions, dermatitis pruritus and joint pain; anorexia, nausea, vomiting, gastrointestinal upset, abdominal pain; fever, malaise, headache, and dizziness; mental confusion, disorientation and possible hallucinations. Numbness and tingling of the extremities due to peripheral neuritis have been reported infrequently.

Elevated serum uric acid levels occurs and precipitation of acute gout has been reported.

Transient impairment of liver function as indicated by abnormal liver function tests is not an unusual finding. Since ethambutol is recommended for therapy in conjunction with one or more other antituberculous drugs, these changes may be related to the concurrent therapy.

Pyrazinamide

Flushing is quite common and hypersensitivity reactions and photosensitization rarely occur. Mild degrees of anorexia and nausea are common, but vomiting is less frequent.

Clinically, the two most important reactions are hepatitis and arthralgia. The frequency of hepatotoxicity declines with reduction of dosage even in combination with isoniazid or Rifampin.

Inhibition of the renal tubular secretion of uric acid by pyrazinoic acid, the main metabolite of the drug, increases the serum uric acid concentration and may precipitate an acute attack in patients with gout. Hyperuricaemia is reduced by co-administration of Rifampin, evidently by facilitation of uric acid excretion rather than any effect on metabolism. Sideroblastic anaemia is a rare, reversible reaction. Convulsions have been described.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Rifampin

Nausea, vomiting, and increasing lethargy appears within a short time after ingestion; unconsciousness may occur when there is severe hepatic disease. Brownish-red or orange discoloration of the skin, urine, sweat, saliva, tears and feces will occur, and its intensity is proportional to the amount ingested. Liver enlargement, possibly with tenderness, may develop within a few hours after severe overdose; jaundice may develop rapidly. Other physical findings remain essentially normal. Bilirubin levels may increase rapidly with severe overdose. The treatment is supportive.

Isoniazid

Nausea, vomiting, dizziness, slurring of speech, blurring of vision, and visual hallucinations are early manifestations of overdose. Respiratory distress and CNS depression, progressing to coma, with or without seizures may occur with marked overdose. Severe metabolic acidosis, acetoneuria, and hyperglycemia are typical laboratory findings.

Treatment consists of gastric lavage following intubation and the control of convulsions by diazepam given intravenously. Large doses of pyridoxine may have to be given intravenously. Metabolic acidosis can be corrected by sodium bicarbonate.

Ethambutol

For Ethambutol overdose, there is no specific antidote, but gastric lavage should be employed to remove the drug from stomach.

Pyrazinamide

Overdose experience with pyrazinamide is limited. In one case of Pyrazinamide overdose, abnormal liver function tests developed, which spontaneously reverted to normal when the drug was stopped.

Treatment of overdose is gastric lavage and supportive therapy.

Effects on ability to drive and use machines

AKurT-4 has minor to moderate influence on the ability to drive and use machines.

Undesirable effects of ethambutol, such as confusion, disorientation, hallucinations, dizziness, malaise and visual disturbances (blurred vision, red-green color blindness, loss of vision) may impair the patient's ability to drive or operate machinery.

PRESENTATION

Available as a pack Alu- Alu of 15 x 6 Tablets.

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